Enhancing Booking.com AI-Powered Innovation for a Seamless Travel Experience

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ABSTRACT

The proposed project brings a strategic innovation to the Booking.com mobile app in the form of an AI-based trip planner that will change the approach to planning, personalization and booking of the traveling experiences. The existing Booking.com version, despite its popularity, has a number of user issues, including inaccurate hotel check-in/out times search, lack of bus ticket and tour guide booking, confusion whether a car is available in a country or not, and a lack of a coherent user experience when a user tries to book a flight, hotel, transport, or attraction. One of the most common problems that users tend to face when traveling extensively or as they complete their first trip as a traveller is that it becomes too complicated to maintain their flight schedules using multiple tools. This project caters to these concerns by incorporating smooth automation, user-friendly universal interface, and feature upgrading in a consolidated AI-assisted mobile app. The enhanced application has a customized intelligent Trip Planner option which allows users to easily build comprehensive itineraries by entering their travel preferences, including destination, duration, budget, and interests. Notable enhancements encompass prominently featured hotel check-in/check-out time information, an exclusive bus tickets reservation page, tour guide programs arranged by languages, and a clearer indication of the presence of car rentals by country. In order to best satisfy user needs, interviews were arranged with hotel representatives, frequent travellers, and potential users, including family and friends, to get a sense of their practical needs and expectations related to usability, which informed much of the final prototype design. The project adopted Agile methodology in the Software Development Life Cycle (SDLC) where an iterative development style was observed as well as regular feedback and continuous enhancement. The agile stages included requirement analysis by consulting the user, interactive system design involving structured diagrams such as Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), and use case diagrams, and iterative prototyping with the help of the design tool Figma. The usability testing sessions involved family and friends and were carried out on each of the iterations, with the testers delivering qualitative data on the navigability of the system, clarity of features, and the effectiveness of user interface design. This feedback enabled the necessary improvement, such as the change in the visibility of itinerary summaries, improvement in the icon size, and streamlining the booking confirmation interface. In terms of business, the improved application can provide Booking.com with competitive advantage using next level of personalization and service bundling, which can lead to higher user engagement and retention. To sum up, this project is successful in illustrating how the ideas of agile in systems analysis and design can considerably improve the user experience by providing real-life solutions to the problem of digital travel planning innovation in AI.

Key Words: AI-Trip Planner, Agile Methodology, User Feedback Evaluation, Smart Itinerary Generation, Multimodal Travel Integration

1. INTRODUCTION

Booking.com was founded in 1996 in Amsterdam. This is an application that allows users to search, compare and book flight tickets, accommodations, car rentals, taxis, attractions and travel experiences. Their aim is to make it easier for everyone to experience the world (Booking.com, n.d.). However, users often encounter certain challenges when planning a trip online. Examples of these are challenges in checking flight and hotel rates on various apps, missing information about hotel check-in and check-out times, no combined support for car rentals and public transport and having different UIs leading to making costly bookings. Using many different apps and websites can not only slow down your journey but also bring risk of miscommunication to travellers. Thus, the aim of enhancement for Booking.com is to be an

AI-based travel planner and hotel reservation service, which will be the major extension of the current Booking.com mobile application.

A study by Accenture in 2024 indicates that the travel industry is being transformed rapidly by generative AI, which allows hyper-personalizing the experience and minimizing the time involved in fragmented planning activities. (Robinson, 2024) This is a trend that concurs with Booking.com desire to provide a single AI assistant that would make trip choices straightforward and enhance customer engagement. Simultaneously, according to Ensemble Travel Advisors, the industry is standing at the dawn of a golden age of demand, in which more tech-savvy travellers are looking to experience a more meaningful, seamless, and efficient travel (Gould, 2024). These reflections highlight the timeliness and significance of designing a coherent, smart travel planning tool, such as the one which is the focus of the present project.

i. Problem Identified

1. Users, especially first-time travellers, face challenge on trip planning.
2. Hotel check-in and check-out times are not well displayed, which can cause confusion.
3. “Tour Guide” section could be added to Booking.com.
4. “Bus Ticket” section could be added to Booking.com.
5. Booking.com does not clearly indicate which countries support the “Car Rental” function.

ii. Milestones

1. Develop a new AI Trip Planner for the user
2. Highlight check-in and check-out times at the top of the hotel information page.
3. Add a “Tour guide” section and mainly sort by the languages
4. Add a “Bus Ticket” section that offer users to choose alternative transportation other than flights and car rentals.
5. Add a country selection dropdown to show which countries is support the “Car Rental” section.

iii. Project Timeline

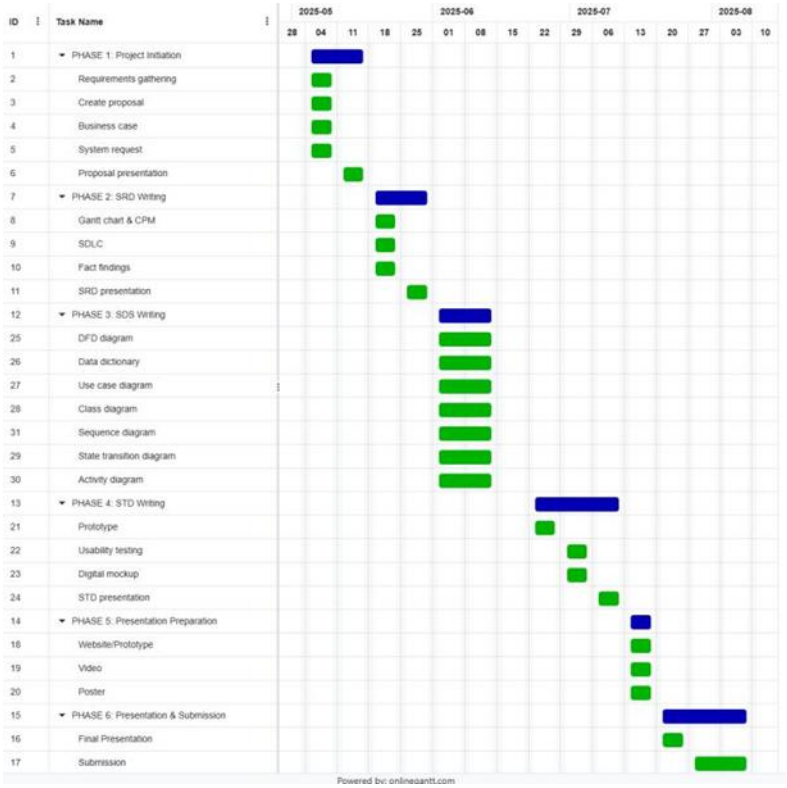


Figure 1: Gantt Chart for the project

iv. Software Development Life Cycle (SDLC)

The software development life cycle (SDLC) is a step-to-step cycle process that help a development team to create high-quality, cost-effective and secure software (Jackson, 2024). This is an important step for Booking.com to develop a complex feature like AI Trip Planner which

require a large amount of money. Thus, one of the key benefits of using SDLC is its ability to manage costs effectively (Leed Software Development, 2023). By clearly outlining the AI Trip Planner scope, requirements, and budget from the beginning, Booking.com can prevent overspending and ensure optimal use of resources.

1. Requirement Analysis

The team gathers and analyses business requirements such as users, usage, data input, and expected output to create a clear specification document that guides the entire development process (Saigon Technology, 2024). Based on our research, we found some additional features that improve user experience. Thus, project proposal and business case was created to guides us until the end of the project.

2. System Design

During this step, we need to use tools to visualize information about the application's behaviour and structure (GitHub, 2025). Thus, we had made use case diagram, activity diagram and state transition diagram to know how user will interact with the system. Besides that, we made class diagram, entity relationship diagram and DFD diagram to design the database system.

3. Implementation

Implementation step involves writing code, building software components, and integrating them to create a functional product (Kamboj, 2025). For our project, we had selected Figma as our platform to design and build the AI Trip planner, tour guide and some small improvements.

4. Testing

Testing steps need to ensure no bugs or error occurred by fixed the issue before moving to deployment section (Coursera Staff, 2025). We had conducted the usability testing ourselves. After that, we invited some people to do pilot testing and usability testing to find any bugs in our result to fix.

5. Deployment

After we fixed all the bugs from the testing part, we can officially launch our new feature to the Booking.com application. Thus, all users can use the new feature.

6. Maintenance

After the deployment of the new feature, maintenance is important where it ensures that the software remains good and follow the trend (Atlassian, n.d.). For example, the new AI model is release so it can replace the previous AI model.

2. RELATED STUDIES

In order to get an idea of exactly what current travel websites offer in terms of digital way of planning and personalizing trips, the paper looked at some of the most relevant systems that represent different rates of artificial intelligence (AI) incorporation, usability aspects, and reservations options. Examples worth mentioning are Google Travel, Expedia, Kayak, GuideGeek and AirAsia MOVE, and together they reveal the changing dynamics of travel application styling.

Besides, Google Travel has a focus on inspiring the user to the crafting of their travel itineraries using AI. By typing their entry in natural language such as "7-day trip in Japan" users will see a whole suggested itinerary with maps, photos, and reviews. Nonetheless, Google does not support direct booking in its ecosystem, a factor that creates discontinuity in the user experience by redirecting them to third-party apps (McHugh-Johnson, 2025). Comparatively, Expedia has a ChatGPT-enabled assistant that integrates natural conversation into the app so that users can get real-time hotel, activity and trip options based on preferences, and these suggestions directly link into the booking process in the app (Reuters, 2023). This natural-language interface also allows the user to move between inspiration and action within the app.

Kayak has its own corpus of chat logs of billions of travel queries that it uses to train generative AI that powers its Ask KAYAK search tool where travellers can conduct natural-language queries like family vacation under RM1500 within 3 hours of KL. Although Kayak is mainly a metasearch platform and directs bookings to other websites, its AI performance is timely and easy to use (Kayak, 2024). In the meantime,

GuideGeek, a product created by Matador Network, offers AI-based travel recommendations in the form of messaging features such as WhatsApp and Facebook Messenger. With over 1,000 travel integrations and powered by GPT-4, GuideGeek is more personalized than transactional and can provide a destination overview and perfectly personalized trip ideas but does not enable directly made reservations (Matador Network, 2024).

AirAsia MOVE is a travel superapp designed to book flights, hotel reservations, use a ride-sharing service, duty-free products, and dining in a single platform (AirAsia, n.d.). It allows integrating across services and loyalty options through the BigPay and AirAsia Rewards. Nevertheless, itinerary customization based on AI or voice-supported trip planners are not available yet, which restricts its flexibility in custom trip scheduling. Likewise, Singapore Airlines mobile application includes such features as natural language processing, image recognition, real-time translation, and baggage measurement tools (Singapore Airlines, 2019). These steps help increase traveller independence and minimize the friction of the pre-departure process, though the personalization of the system itself is more tool-oriented than trip-based.

Overall, though, none of these systems have fully brought together AI trip planning, real-time spoken support, and direct multi-service reservations into a single application. The proposed Booking.com AI Trip Planner is unique in its aspiration to fill this gap, achieve the intelligence of itineraries of Google, the booking capabilities of Expedia, and having the conversation interface of GuideGeek, and being highly accessible to the range of users be it tech-savvy or inexperienced. Therefore, our enhanced Booking.com solution fills a crucial market gap by consolidating intelligent trip planning, voice-based interaction, and seamless in-app booking features previously only available in fragmented form across competing platforms

3. METHODOLOGY

i. Data Dictionary

User Database

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
UserID	Unique identifier for user	Integer	N/A	Primary Key, Auto-increment	N/A	Booking (One to Many)
FirstName	First name of user	String	30	Not Null	N/A	N/A
LastName	LastName of User	String	30	Not Null	N/A	N/A
Email	Email address of user	String	80	Not Null, Unique	N/A	N/A
PhoneNumber	Phone number of User	Integer	20	Not Null	N/A	N/A
City	Place of user	String	100	Not Null	N/A	N/A

Hotel Database

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
HotelID	Unique identifier for hotel	Integer	N/A	Primary Key, Auto-increment	N/A	One to Many (Booking)
HotelName	Name of hotel	String	80	Not Null	N/A	N/A
Location	City or Area of hotel	String	100	Not Null	N/A	N/A
PricePerNight	Rate per night	Decimal	N/A	Not Null	0.00	N/A
MaxGuest	Max number of guest	Integer	N/A	Not Null	N/A	N/A
Rating	Guest rating	Decimal	N/A	Nullable	Null	N/A
Description	Hotel facilities description	String	N/A	Nullable	Null	N/A

Bus Database

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
BusID	Unique identifier for bus	Integer	N/A	Primary Key, Auto-increment	N/A	One to Many (Booking)
Name	Name of bus company	String	100	Not Null	N/A	N/A
FromLocation	Departure City	String	100	Not Null	N/A	N/A
ToLocation	Destination city	String	100	Not Null	N/A	N/A
DepartureTime	Bus departure time	Datetime	N/A	Not Null	N/A	N/A
ArrivalTime	Bus arrival time	Datetime	N/A	Not Null	N/A	N/A
Price	Ticket price	Decimal	N/A	Not Null	0.00	N/A
SeatsAvailability	Available seats	Integer	N/A	Not Null	N/A	N/A

Booking Database

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
BookingID	Unique identifier for each Booking details	Integer	N/A	Primary Key, Auto-increment	N/A	One to One (Payment)
BookingType	Selected type of booking (hotels, flights, cars, etc.)	String	30	Not Null	N/A	N/A
BookingStatus	Status of the booking	String	30	Not Null	Pending	N/A
PaymentStatus	Whether the payment has been made	String	N/A	Not Null	Unpaid	N/A
ConfirmedAT	Date and time the booking was made	DateTime	N/A	Not Null	N/A	N/A
Duration	Duration of the booking	DateTime	N/A	Not Null	N/A	N/A
TotalAmount	Total cost of booking	Decimal	N/A	Not Null	0.00	N/A
UserID	ID of the user who made the booking	Integer	N/A	Foreign Key (User)	N/A	Many to one (User)
HotelID	FK to Hotel table	Integer	N/A	Foreign Key (Hotel)	NULL	One to One (Hotel)

Flight Database

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
FlightID	Unique identifier for flights	Integer	N/A	Primary Key, Auto-increment	N/A	One to One (Booking)
AirlineName	Name of Airline	String	100	Not Null	N/A	N/A
DepartureCity	City of departure	String	100	Not Null	N/A	N/A
ArrivalCity	Destination city	String	100	Not Null	N/A	N/A
DepartureTime	Flight departure time	Datetime	N/A	Not Null	N/A	N/A
ArrivalTime	Flight arrival time	Datetime	N/A	Not Null	N/A	N/A
Price	Ticket cost	Decimal	N/A	Not Null	0.00	N/A
SeatAvailability	Available seats	Integer	N/A	Not Null	0	N/A

Tour Guide Database

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
GuideID	Unique identifier for tour guide	Integer	N/A	Primary Key, Auto-increment	N/A	One to One (Booking)
Name	Name of guide	String	100	Not Null	N/A	N/A
Language	Languages spoken	String	100	Not Null	N/A	N/A
TourType	Tour Categories	String	100	Not Null	N/A	N/A
Rating	Average user rating	Decimal	N/A	Nullable	Null	N/A
PricePerDay	Cost per day of service	Decimal	N/A	Not Null	0.00	N/A
Description	Tour Guide profile	String	100	Nullable	Null	N/A

Payment Database

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
PaymentID	Unique identifier for the payment	Integer	N/A	Primary Key, Auto-increment	N/A	One to One (Booking)
BookingID	Asscociated booking being paid for	Integer	N/A	Foreign Key (Booking)	N/A	One to One (each booking once)
TotalAmount	Total cost paid	Decimal	N/A	Not Null	0.00	N/A
PaymentMethod	Method used (Creditcard, Debitcard, etc.)	String	50	Not Null	N/A	N/A
Status	Payment status	String	20	Not Null	Pending	N/A
PaidDuring	Timestamp of successful payment	Datetime	N/A	Not Null	N/A	N/A
UserID	Bill for the user	int	N/A	Not Null	N/A	One to One (user)

RentalCar Database

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
CarRentalID	Unique identifier for car	Integer	N/A	Primary Key, Auto-increment	N/A	One to One (Booking)
CompanyName	Car Rental company name	String	100	Not Null	N/A	N/A
CarModel	Model of car	String	100	Not Null	N/A	N/A
Location	Pickup Location	String	100	Not Null	N/A	N/A
PricePerDay	Daily rental cost	Decimal	N/A	Not Null	0.00	N/A
Availability	Availability status	Boolean	N/A	Not Null	True	N/A

TripPlannerAI Database

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
PlanID	Unique identifier for each Plan	Integer	N/A	Primary Key, Auto-increment	N/A	One to Many (Booking)
UserID	Unique identifier for user	Intteger	N/A	Not Null	N/A	Many to One (Booking)
Budget	User's budget preference	Decimal	N/A	Not Null	0.00	N/A
DescriptionText	User input preference trip details	String	200	Not Null	N/A	N/A
Duration	Duration preference of trip	Datetime	N/A	Not Null	N/A	N/A
Destination	Destination set by user	String	100	Not Null	N/A	N/A
CreatedAt	Timestamp of when the plan was created	Datetime	N/A	Not Null	N/A	N/A
UserID	ID of the user who made the booking	Integer	N/A	Not Null	N/A	Many to one (User)
HotelID	FK to Hotel table	Integer	N/A	Foreign Key (Hotel)	NULL	One to One (Hotel)S
FlightID	FK to Flight table	Integer	N/A	Foreign Key (Flight)	NULL	One to One (Flight)
BusID	FK to Bus table	Integer	N/A	Foreign Key (Bus)	NULL	One to One (Bus)
RentalCarID	FK to RentalCar table	Integer	N/A	Foreign Key (RentalCar)	NULL	One to One (RentalCar)
TourGuideID	FK to TourGuide table	Integer	N/A	Foreign Key (TourGuide)	NULL	One to One (TourGuide)
PaymentID	FK to Payment table	Integer	N/A	Foreign Key (Payment)	NULL	One to One (Payment)

Figure 2: Data Dictionaries

ii. Class Diagram

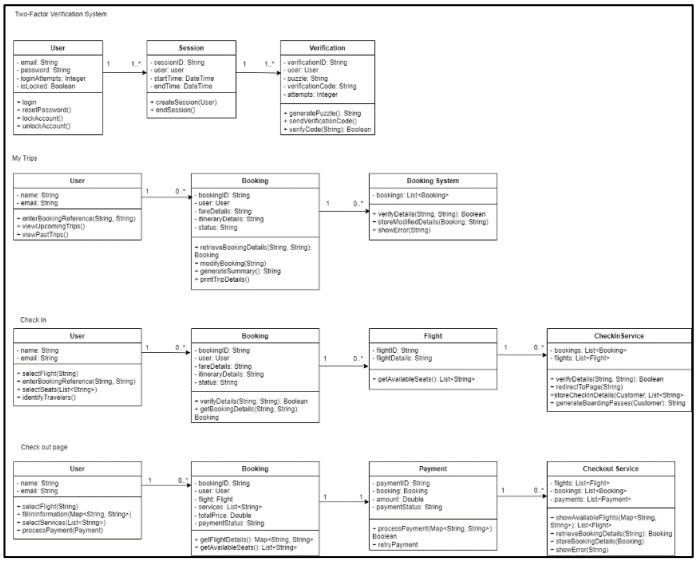


Figure 3: Class Diagram

The user class represents the user interaction with the booking system. It stores personal information like the userID, name, email, and phone number. The user can make multiple types of bookings through the booking class where it is a one-to-many relationship. The user class also enables the users to register, login, and create bookings according to their choice itself.

Then, the admin class controls the whole system where it allows the admin to access rights to manage the system. Although the admin class shares a common action with the user class, an admin can manage multiple users and bookings implying it as a one-to-many relationship that allows them to oversee all the system operations.

The booking class is the main class that allows users to book and confirm their reservation. The class stores information like the booking status for each separate service and it is associated with one payment window. The booking class also has a one-to-one relationship with the services available to allow each user to book what they prefer easily. Additionally, each Booking is also associated one-to-one with each service class (Hotel, Flight, Bus, TourGuide, RentalCar), ensuring clear tracking of which services a user has selected.

The payment class stores all the booking information and amount of the booking that is prepared by the booking class. It records all the financial transactions details for a booking that includes the paymentID, amount of the booking, paymentMethod, paymentStatus, and the bookingID. Payment is directed by one booking signifying a one-to-one relationship with the booking class.

The AI Trip Planner class automates travel planning with attributes like destination, budget, and preferences. The AI Trip Planner uses these inputs to recommend bookings and generate complete travel plans. It connects directly to the Booking class, allowing users to transform an itinerary into confirmed bookings. It also maintains an association with the User class (one-to-many), where each user can create and manage multiple AI-generated plans.

The services that the system provides are hotel, bus, tour guide, flight, and car rental bookings. Each of the class contains specific information for each service that allows the users to book their own preference. The information on the booking of each service will then be stored back to the booking class to create the final booking information of the user and will display the amount of the booking for the user.

iii. Sequence Diagram

Sequence Diagram 1 – AI Trip Planner

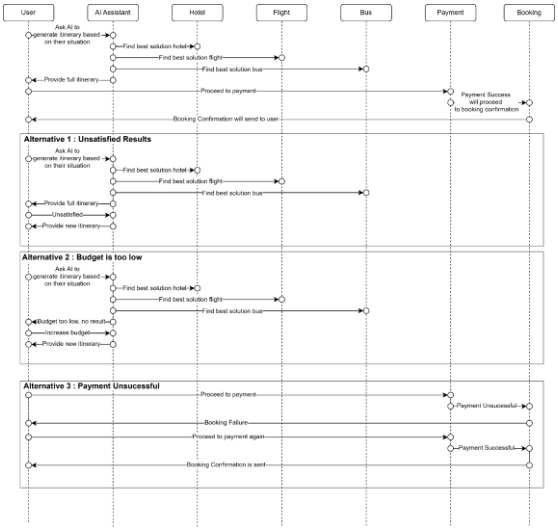


Figure 4: Sequence Diagram 1 – AI Trip Planner

The process begins when the user asks the AI assistant to generate a personalized travel itinerary based on their situation such as preferences, budget, or timing. The AI assistant will help to search for hotels, flights, and buses to find the best possible combinations and show a full itinerary. If the user is satisfied, the user will proceed with the payment to book them. After the payment is successful, the system confirms the booking and sends confirmation to the user. However, if the user is unsatisfied with the itinerary, the AI assistant will search and provide a new itinerary. In cases where the user's budget is too low to find any suitable travel plans, the AI notifies the user to adjust their budget, then AI provides new searches. If the payment process fails, the system informs the user of the booking failure and attempts the payment again until

successful, after which the booking is confirmed and sent to the user.

Sequence Diagram 2 – Manual Booking

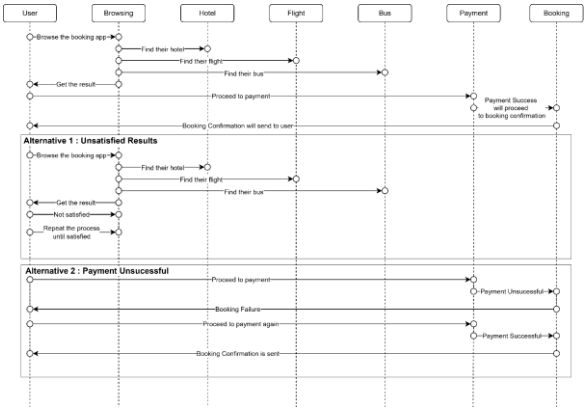


Figure 5: Sequence Diagram 2 – Manual Booking

The user begins by browsing property, searching for flights, bus or other services to book for their vacation. When users are satisfied with the result, they can proceed to their payment via multiple options. When the payment is successful, the system will complete their booking and confirmation page will be sent to the user. If the users are unsatisfied with their result, they can return to browsing listing to continue their search until they find an alternative that meets their requirements and satisfaction. If the payment fails, the system will inform the user with a notification stated that the booking attempt fails and require users to retry the payment until it is successful, then the booking is confirmed, and a notification will be sent to the user.

Sequence Diagram 3 – Transportation Booking

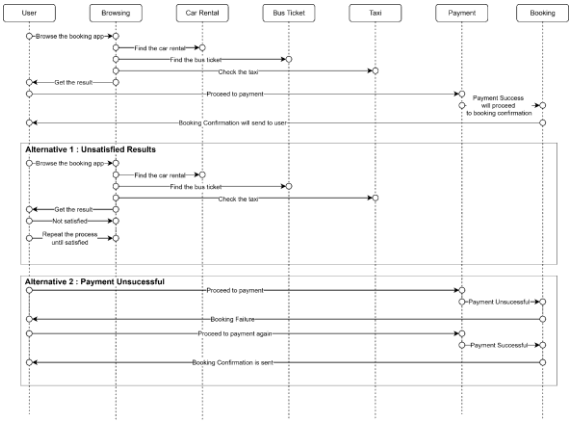


Figure 6: Sequence Diagram 3 – Transportation Booking

The use case starts with the user using the booking application. The browsing component within the system initiates the search of the services, one at a time by querying the car rental, bus ticket, and taxi modules. After retrieving all the results of the services, the results are sent back to the user to view the same. In case of customer satisfaction with the variety provided, he or she moves to payment. When the payment module returns a successful payment the booking system will create a booking confirmation and send it back to the user thus completing the transaction.

There are also two additional flows shown in the diagram. With Alternative 1 (Unsatisfied Results), the user explores the app and obtains the results of transportation searches. Nevertheless, in case the user fails to be content with the result, he or she repeats the process of browsing. This loop is repeated until it comes up with satisfactory results upon which the user would proceed to payment. In Alternative 2 (Payment Unsuccessful), the user clicks on payment when they have already chosen their favoured services but the first attempt to pay does not work. When the system detects a failure, the user is informed, and he repeats the payment. The second time it works out, and a booking confirmation is created, which is passed to the user.

Sequence Diagram 4 – Tour Guide Booking

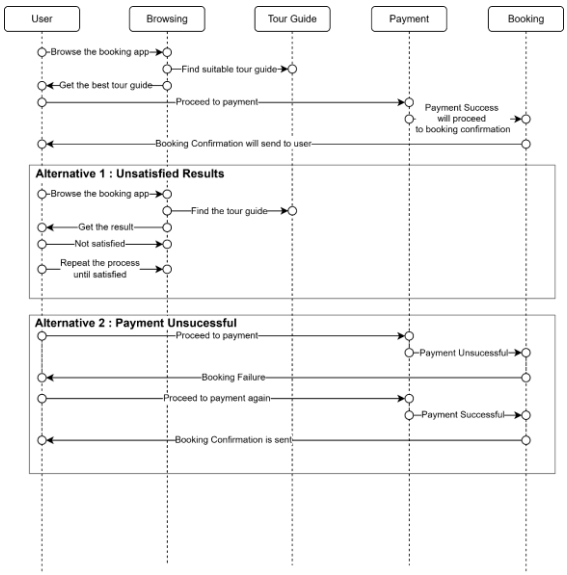


Figure 7: Sequence Diagram 4 – Tour Guide Booking

The user started to browse Booking.com application to look for a suitable tour guide to bring them play around a specific city. If they have found the best tour guide to choose, they will proceed to book and make their payment. After the payment is successful, the booking confirmation will be sent to the user. If they do not find the satisfied tour guide, they will continue to seek the best tour guide that fulfils their requirements. If their payment is unsuccessful, the system will ask user to make the payment again until the payment is successful.

iv. State Transition Diagram

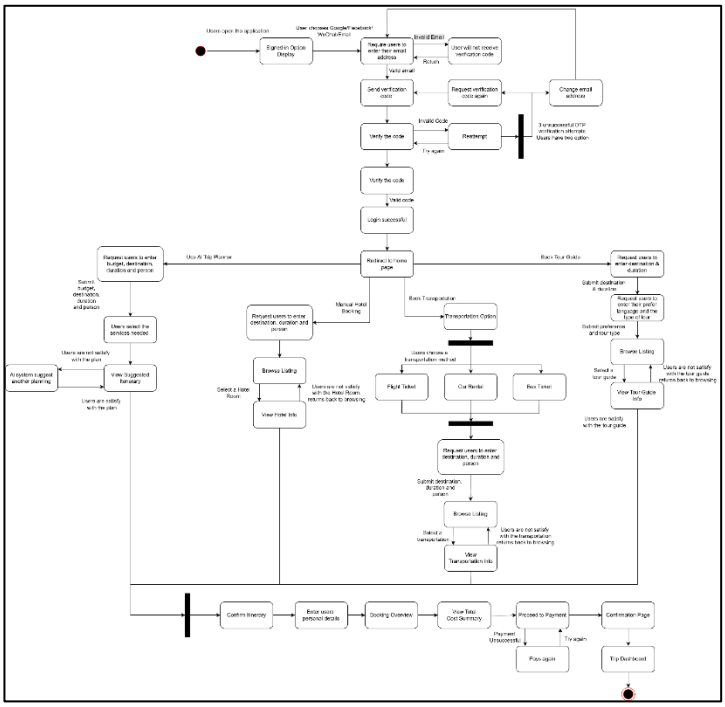


Figure 8: State Transition Diagram

The Initial step of the system requires users to log in into Booking.com application. Our system provides various login options, which include Google, Facebook, WeChat or Email Address. One of the most used methods is email address. In this process, users must enter their email address to receive a verification code. The verification code will be sent strictly to the email address entered. Therefore, users

must ensure to enter their address correctly, or else users would need to re-enter their valid address. Once users receive and enter the code, the system will verify it. If the code is valid, login will be successful. If the code is invalid, users are allowed to retry up to three attempts. After the third failed attempt, the system will reject the current code, and users will be given two options which are request a new verification code or return to the email input step to provide a different email address.

After they log in successful, the users will be redirected to the home page. If they want to make a plan but not sure how to begin, they can use the AI-Driven Trip Planner feature to help them plan a full itinerary and package. The system will request users to enter their budget, destination, duration and number of travellers for the trip. Next, users could select the service they needed for the trip such as a full itinerary (including flight, hotel, transportation booking and tour guides booking), hotel booking, flight booking and/or bus transportation. Once services are selected, the AI system will provide a suggestion based on user's requirement, if users are not satisfied with the suggestion, they can request for a new plan until users are satisfied with the suggestion overview.

If users want to book a hotel room, they can use the manual hotel booking system. The system will prompt users to enter their destination, duration and number of travellers for the trip. After filling in the request, a list of relevant hotels will be display for users to browse. Users can view each hotel's details for more information. If they are not satisfied with a particular option, they can return back to the list and continue browsing until they find a suitable hotel.

If users want to book transportation, they can choose from three service which are flight, car rental and/or bus ticket. Once a service is selected, the system will prompt users to enter their destination, duration and number of travellers for the trip. After filling in the request, a list of relevant flight/ car rental/ bus option will be display for users to browse. Users can view for detailed information about each option. If they are not satisfied with a particular option, they can return to the list and continue browsing until they find a suitable transportation option.

Users can use the tour guide booking system to find a suitable tour guide for their trip. When the system is selected, the system will prompt users to enter their destination, duration, prefer language and the type of tour for the trip. After filling the request, a list of relevant tour guide will be display for users to browse. Users can view for detailed information about each guide including their rating, review and other relevant details. If users are not satisfied with a particular option, they can return to the list and continue browsing until they find a suitable tour guide.

When satisfied, users can confirm the itinerary. At this stage, the system will prompt users to enter their personal details to ensure the booking is associated with related person. Users can then review all details in the Booking Overview section before proceeding to payment. If the payment fails, the system will prompt users to retry until it is successfully processed. Once the payment is completed, the system will redirect users to a confirmation page, where they can view and manage their trip details via the Trip Dashboard.

v. **Activity Diagram**

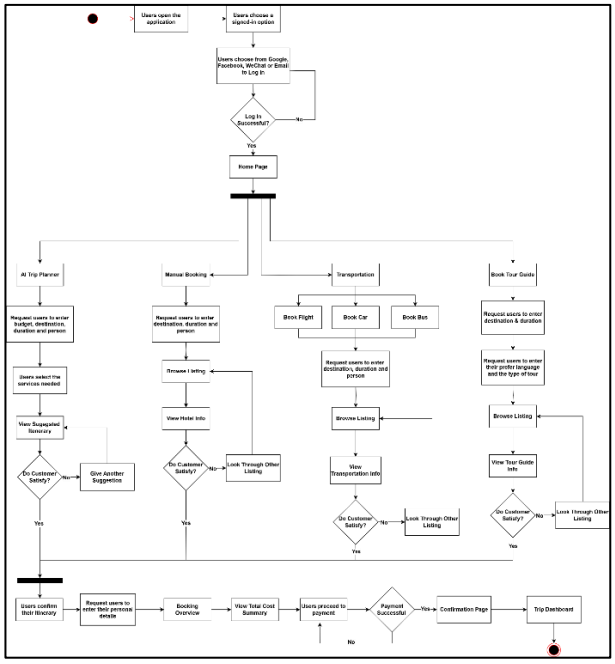


Figure 9: Activity Diagram

The Initial step of the system requires users to log in into Booking.com application. Our system provides various login options, which include Google, Facebook, WeChat or Email Address. If the log in was not successful, the system will request users to try again.

Upon successful login, users are directed to the home page. If users are planning for a vacation but don't know where to start, they can try out the AI-Driven Trip Planner feature for help. To begin with, the system will ask users to provide details of their trip such as their budget, destination, duration and number of travellers. Next, users can choose different services they would require during the trip such as a complete itinerary (including flight, hotel, transportation, and tour guide bookings), individual hotel reservation, individual flight booking, and/or individual bus transportation. Once services are selected, the AI system will generate a suggestion based on the user's needs. If users find this suggestion unsatisfactory, they can request a new plan until they are content with the suggestion overview.

For booking a hotel room, users can utilize the manual hotel booking system. Users will be prompted by the system to provide their destination, trip duration, and number of travellers. Once users have completed the request, a list of pertinent hotels will be shown for them to browse. Users can access each hotel's details for further information. If they don't find the current hotel to be satisfied, they can go back to the list and keep browsing until they locate an appropriate hotel.

If users want to book transportation, they can choose from three service which are flight, car rental and/or bus ticket. The system will prompt users to enter their destination, duration and number of travellers for the trip. After filling in the request, a list of relevant flight/ car rental/ bus option will be displayed for users to browse. Users can view for detailed information about each option. If they are not satisfied with the current option, they can return to the list and continue browsing until they find a suitable transportation option.

The tour guide booking system allows users to find a suitable tour guide for their trip. When the system is selected, the system will prompt users to enter their destination, duration, prefer language and the type of tour for the trip. After filling the requirements, a list of relevant tour guides will be shown to the users to browse. Users can access the tour guide detail information such as their rating, reviews, and other pertinent details. If users are dissatisfied with a specific option, they can go back to the list and keep browsing until they locate a suitable tour guide.

When satisfied, users can confirm the itinerary. At this stage, the system will prompt users to enter their personal details to ensure that the booking is linked to the corresponding individual. Before proceeding with payment, users can review their details in the Booking Overview section again to ensure no mistakes are made. If the payment fails, users will receive a notification stated that the booking attempt fails and users need to retry the payment until it is successfully processed. Upon payment completion,

users will be redirected to a confirmation page by the system, where they can view and manage their trip details through the Trip Dashboard.

vi. **DFD Diagram**

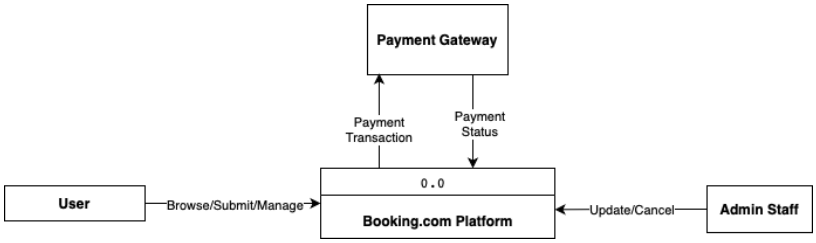


Figure 10: DFD Level 0 Diagram

This is a context diagram of the communications between the Booking.com Platform (Process 0.0) and three external entities, the User, the Payment Gateway and the Admin Staff. The users use the platform to make browsing, submissions, and manage their bookings and the platform interacts with the Payment Gateway through relaying payment transactions and retrieving the payment status confirmations. Admin Staff use the system to correct records of booking or cancel listing. The diagram is at a conceptual level of the main external data exchanges of the system and forms the basis of more detailed process modelling.

Level 1 DFD – Decomposition of Booking.com Platform (Process 0)

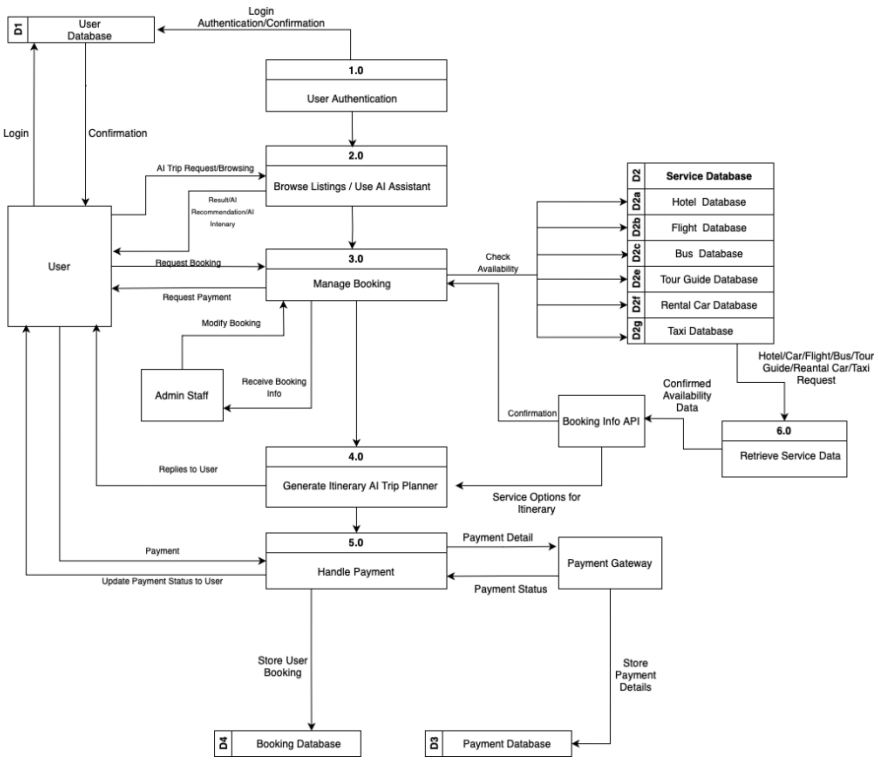


Figure 11: DFD Level 1 Diagram

The Level 1 Data Flow Diagram (DFD) of the Booking.com Platform gives an elaborate breakdown of the major process of the system (Process 0) into 6 sub processes connected with each other. It shows the interaction of the system with other external entities, data stores and other components of the system to provide an improved experience of travel planning and booking. The flowchart indicates the collaboration of the AI assisted planning of itineraries and unified travel experiences in the form of booking hotel rooms, traveling commutes and tour arrangements.

Process 1.0 starts with user authentication. In case a user tries to log on, the user database (D1) is consulted to determine whether the credentials entered by the user are correct. On successful

authentication, the user is then notified on successful authentication and proceeds to access the core features of the application. After the success of the authentication process, the users will be able to search the listings or order AI-made itineraries through Process 2.0, ""Browse Listings / Use AI Assistant," Through this process, the app grasps the preferences of the user and either displays them manually, or calls the AI engine, to give personalized recommendations on travel. The findings in form of accommodation suggestions to complete schedules are transmitted back to the user interface.

Once a user has decided that they are ready, they request a booking using Process 3.0, "Manage Booking." This one deals with receipt of booking information and verifies available services of hotels, flights, buses, tours and car rents among others. Availability requests are communicated to Process 6.0, Retrieve Service Data, which communicates with a set of so-called service databases (D2a-D2f), including Hotel Database, Tour Guide Database, and Rental Car Database, instead of querying each of these databases individually. The availability data is confirmed and sent to the Booking Info API which is then used in the itinerary or during the booking confirmation.

The process 4.0, Generate Itinerary AI Trip Planner, takes the confirmed availability information and creates an intelligent travel plan based on the preferences of the user. The remaining plan is posted back to the customer and the entire travel planning service is automated and customized. In case the user decides to accept the itinerary or create any reservation, process 5.0, the process of handling payment is executed. This step transfers details of payment through to the external Payment Gateway and returns the transaction pending status which is then updated in the system. Achieved payment causes data to be inserted into Booking Database (D3) and Payment Database (D4) and notifies the user about the result.

The system also facilitates communication with Admin Staff. Admin Staff can get the information that shows booking and do the necessary change, or corrections unlike in the past with Process 3.0 which gives room to operational flexibility. All the flows of the data are labelled correctly and are routed correctly through system processes and are not directly accessible to users and external systems- compliant to the DFD rules and best practices.

In general, this Level 1 DFD represents a modular view, logically structured version of the improved Booking.com. It demonstrates explicitly how so-called user interactions, back-end service integration, and AI based features could combine to provide a smooth travel planning process. The diagram helps in traceability, scaling of the system and easier comprehension of the connection between the components that make the system achieve its user and administrative functionalities.

4. RESULTS

Login Page

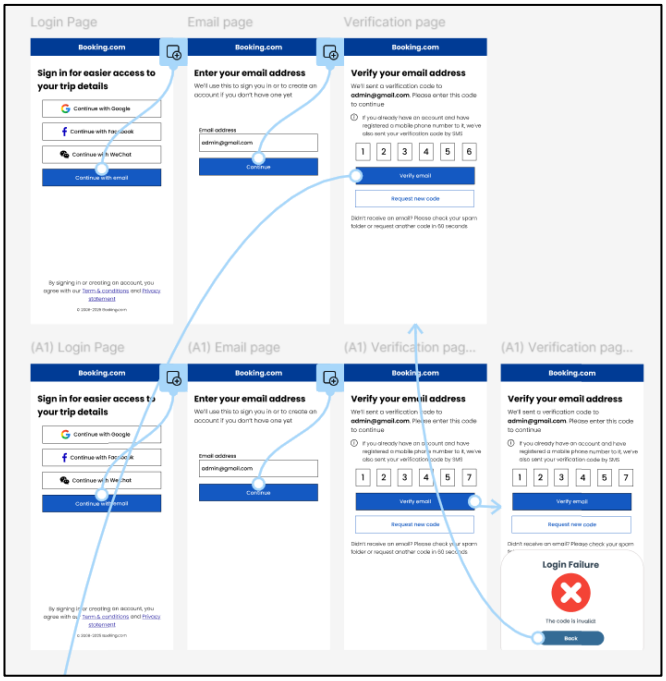


Figure 12: Login Page

Home Page

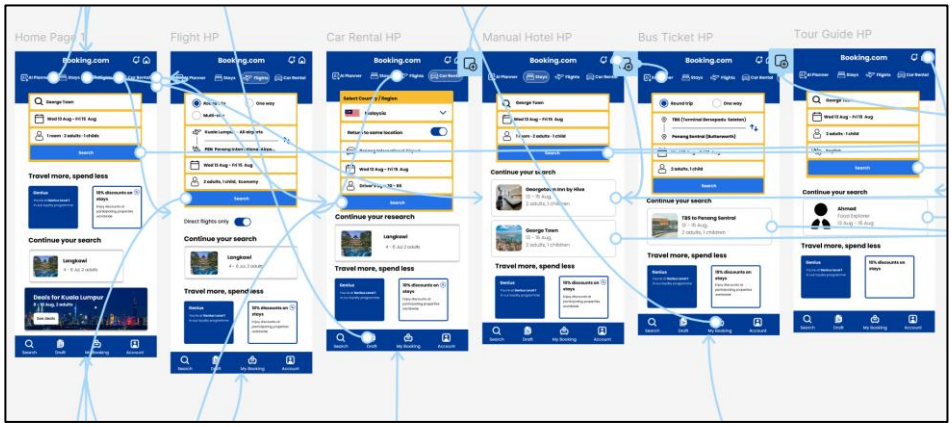
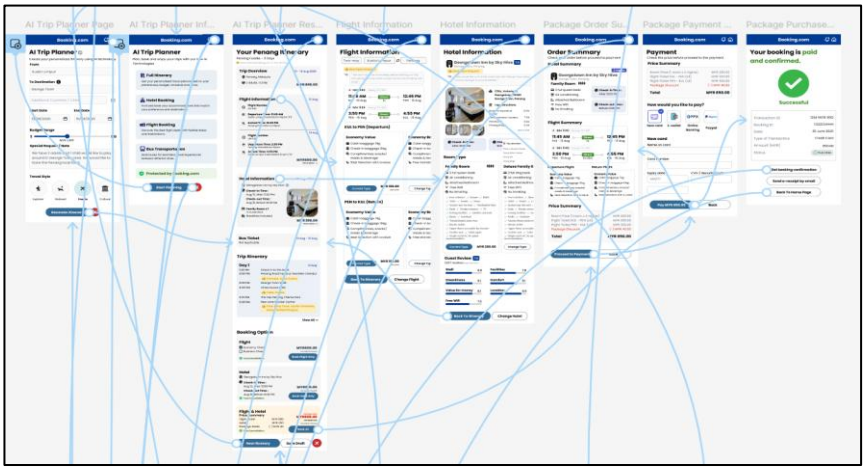


Figure 13: Home Page

Our evaluators tested the Login Page via Email address and Home Page. In the feedback form, we receive a positive response stated the ease of understand on using these interfaces.

AI Trip Planner



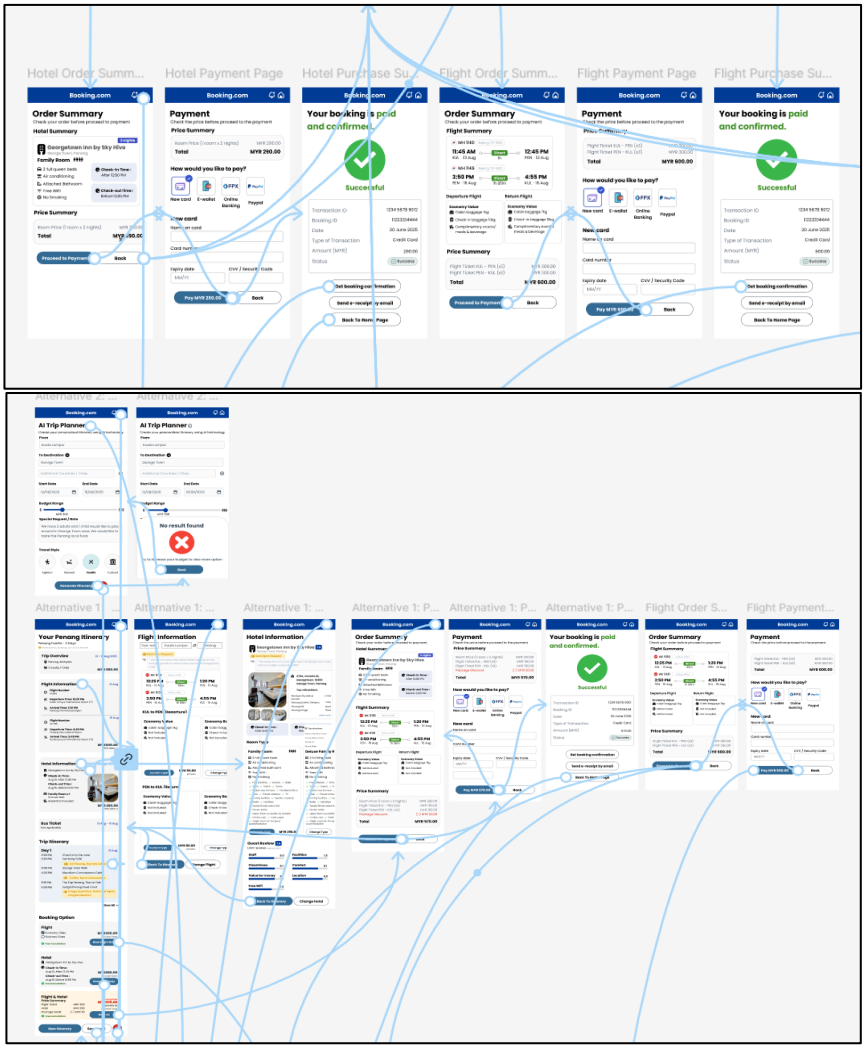


Figure 14: AI Trip Planner

Our evaluators tested the usability of the AI Trip Planner Page. We received positive feedback from users stated a details description of each page, giving sufficient information for evaluators to understand the journey throughout the trip planning process.

Draft

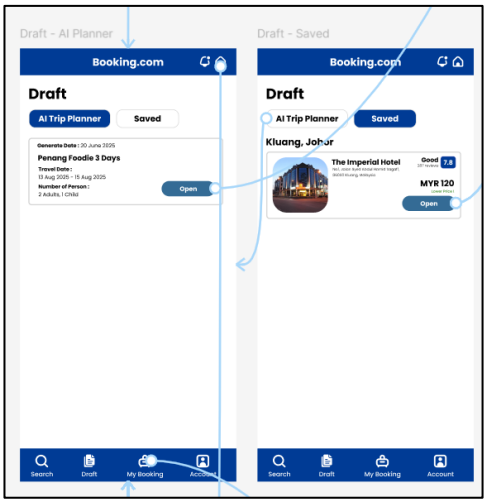


Figure 15: Draft Page

In this segment, our evaluators specifically tested the Draft Page. Our evaluation form received positive response stated easy to navigate.

Flight Booking Flow

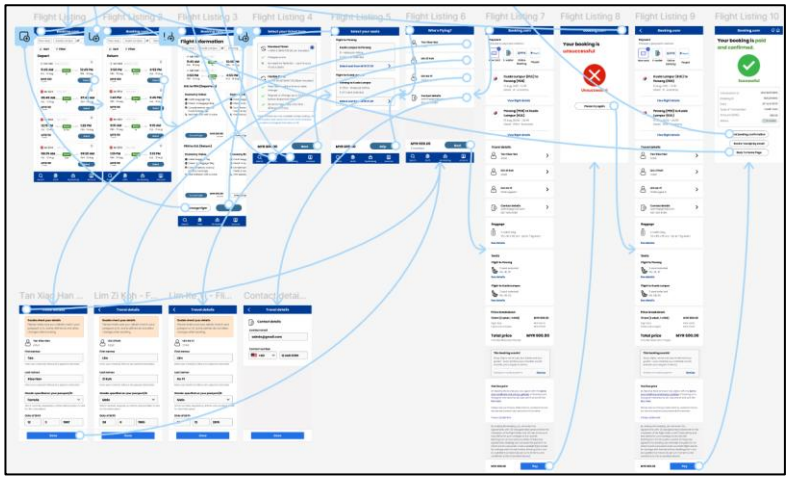


Figure 16: Flight Booking Flow

In this segment, our evaluators specifically tested the Flight Booking Page. Our evaluation form received a positive response. With feedback states a smooth user flow. A well-performed prototype in handling login failures and successful recovery, allowing users to retry and proceed.

Hotel Booking Flow

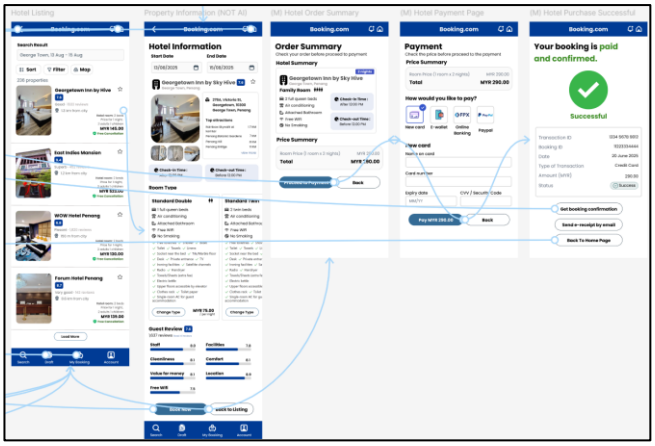


Figure 17: Hotel Booking Flow

In this segment, our evaluators specifically tested the Hotel Booking Page. Our evaluation form received a positive response. With feedback states a smooth user flow. A well-performed prototype in handling login failures and successful recovery, allowing users to retry and proceed.

Car Rental Flow

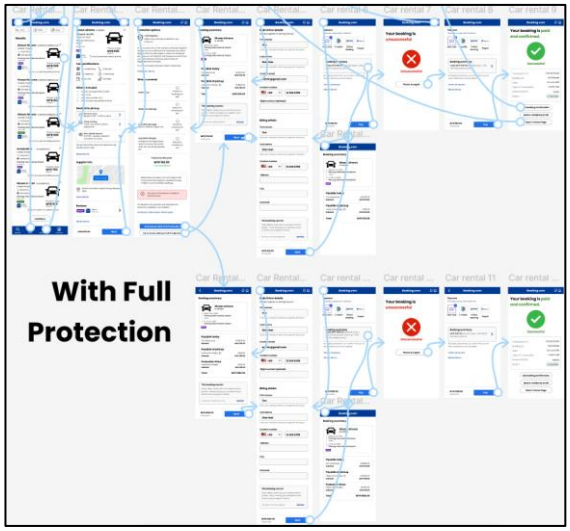


Figure 18: Car Rental Booking Flow

In this segment, our evaluators specifically tested the Car Rental Page. Our evaluation form received a positive response. With feedback states a smooth user flow for with and without the insurance for protection. A well-performed prototype in handling login failures and successful recovery, allowing users to retry and proceed.

Bus Ticket Flow

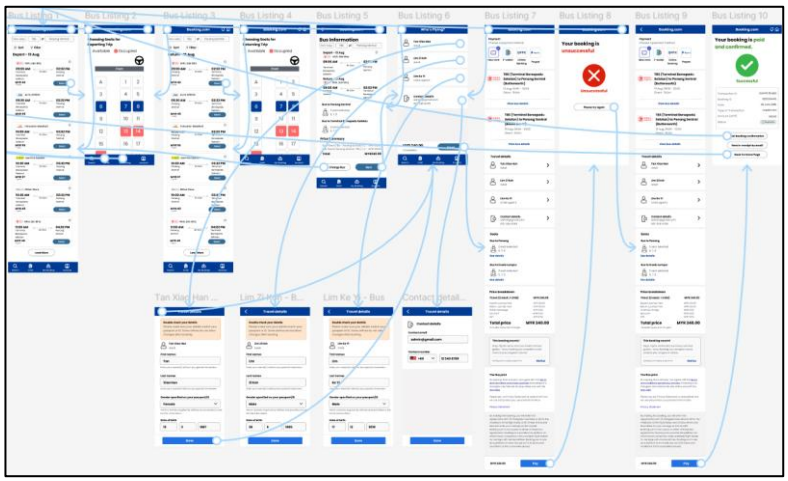


Figure 19: Bus Ticket Booking Flow

In this segment, our evaluators specifically tested the Bus Ticket Page. Our evaluation form received a positive response. With feedback states a smooth user flow. A well-performed prototype in handling login failures and successful recovery, allowing users to retry and proceed.

Tour Guide Flow



Figure 20: Tour Guide Booking Flow

Alternative Tour Guide Flow (Not satisfy)

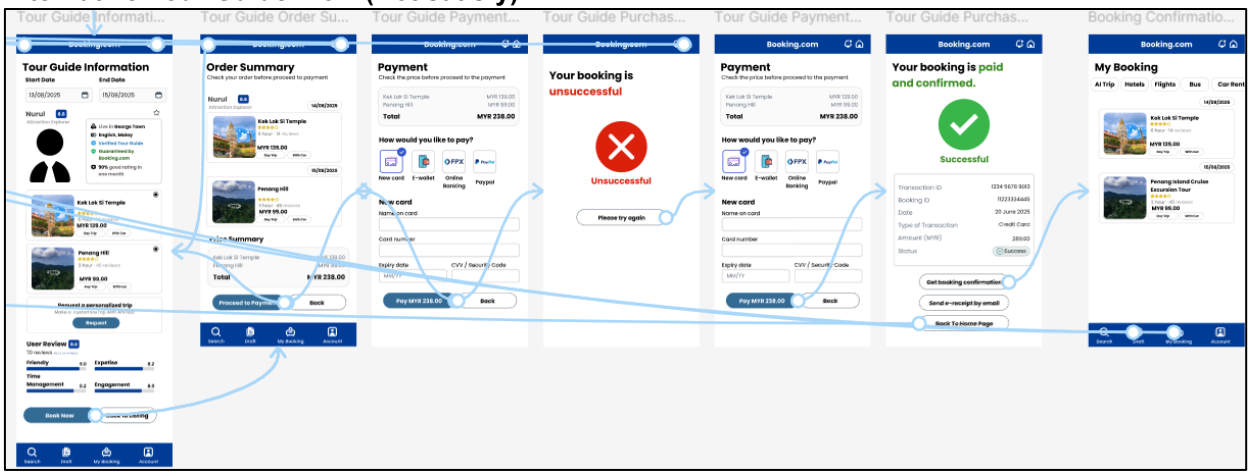


Figure 21: Tour Guide Booking Flow (Alternative)

In this segment, our evaluators specifically tested the Tour Guide Page. Our evaluation form received a positive response. With feedback states a smooth user flow on both normal flow and alternative flow. A well-performed prototype in handling login failures and successful recovery, allowing users to retry and proceed.

Booking Confirmation Page

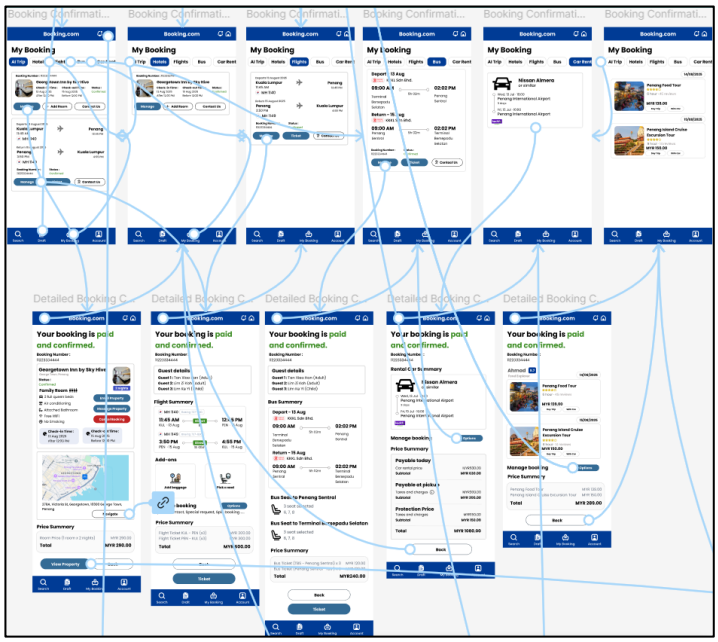


Figure 22: Booking Confirmation Page

Finally, our evaluators tested the Booking Confirmation Page. Our evaluation form received an overwhelmingly positive response. With feedback highlighted a well-designed UI interface and the ease of navigation between different pages.

5. DISCUSSION

Log in Page and Home Page
The evaluator's feedback indicated that users found the login and home interface easy to use and intuitive, which aligns with our design goal of reducing cognitive load having clear interaction pathways to follow.

AI Trip Planner Page
Overall, feedback noted a clear descriptive helps user to carry through the trip planning process. With results shows that users can meet our team objectives by having enough context without being overloaded. The results suggest that we struck a good balance between descriptiveness and consistency.

Draft Page
Evaluators reported a high ease of navigation, which supports the potential success of the user-cantered design layout and logical flow used in this page. The lack of negative feedback on this page suggests that the design decisions of keeping the save state open and formatting the action buttons correctly were what users expected.

Flight, Hotel, Car Rental, Bus Ticket, and Tour Guide Pages
Evaluators across all booking pages reported a smooth user journey, with appropriate fail states. For example, evaluators indicated an easy understanding of navigation even when the interface states payment fail and require users to try again. However, we did receive feedback on not finding the icon to save wanted itinerary, which we later add on in out finalize prototype.

Booking Confirmation Page
We receive positive feedback on this page reiterated the sensibly organized user interface and a seemingly effortful navigation across pages. For example, evaluators appreciated the consistence of the interface, rich details and outline of the booking overall. This captures the design intention to reduce the uncertainty after the booking had occurred. The results indicated a likely of interface UI where users can both go through task bar and read details description.

In general, user response was mostly positive with the test participants having a definite preference for our changes. The tests included key areas such as the Log-In Page, AI Trip Planner, Draft Page, Flight Page, Hotel Page, Car Rental Page, Bus Ticket Page, Tour Guide Pages and Booking Confirmation Page, all which were well-liked by our evaluators.

6. CONCLUSION

This project proposes new features to enhance user experience and attract new customers. The most significant feature is the AI trip planner which design to help first-time travellers and elderly users to create a full itinerary with transportation and hotel booking options. Users can conveniently adjust their itineraries by requesting the AI to regenerate new itinerary or by making some minor changes on their own. This significantly reduce the time to plan a trip by their own. In addition, "tour guide" section is added to provide users more options to choose when they are planning their travel. At the same time, it provides opportunity for tour guides to expand their marketplace by publishing their tour package on Booking.com. Moreover, "bus ticket" section provides alternative transportation to user when travelling between cities.

Through this project, we gained practical experience in applying user-centred design and learned about the importance on recognising user needs and feedback in system design Besides that, we also became familiar with the SDLC using agile methodology, which suggest a continuous improvement cycle on the AI trip planner. Firstly, we identified the user requirements by analysing the review. Besides that,

we developed the system design diagram to understand how the system is going to work. Additionally, we started to build the system using Figma. After that, we conducted usability testing to ensure nothing goes wrong about our system. Lastly, the system is officially launched, and the maintenance will be ongoing on to ensure the system is kept updated.

As a recommendation, a voice-based AI Trip planner could be added to improve accessibility especially for elderly traveller and people with visual impairments. This feature would allow users to interact and talk with AI just like human communication, making the trip planning process more intuitive and hands-free. Thus, it helps to improve the AI trip planner function and enhance user experience with Booking.com.

CONTRIBUTION STATEMENT

Name & Contribution		**SRD**	**SDS**	**STD**	**Full Report**
Liang Yrenn Da	**20%**	1. Detailed of Each Use Case 2. Critical Path Analysis	3. Data Dictionary 4. Activity Diagram 5. State Transition Diagram	1. Initial Prototype 2. Final Prototype	1. Result 2. Discussion 3. Presentation
Yeoh Jie Long	**20%**	1. Introduction	1. DFD Diagram	1. Use Case Testing 2 2. Initial Prototype	1. Video 2. Abstract 3. Related Studies
Ooi Jin Yon	**20%**	1. Overall Use Case Diagram 2. Detailed of Each Use Case	1. Use Case Diagram 2. Sequence Diagram	1. Initial Prototype 2. Final Prototype 3. Use Case Testing 3	1. Poster 2. Introduction 3. Conclusion
Wong Xu Liang	**20%**	1. Nonfunctional Requirement 2. Requirement Reviews	1. ERD Diagram 2. Data Dictionary	1. Use Case Testing 1	1. Methodology
Muhammad Hariz Hakimi Bin Mohd Hafiz	**20%**	1. Project Milestones and Deliverables 2. Project Schedules 3. PERT Chart	1. Class Diagram	1. Use Case Testing 4 2. Digital Mock-up	1. Methodology 2. Presentation

Note:

1. Emily Cha Yu Yue (22049043) had deferred her semester after SDS completed.

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